

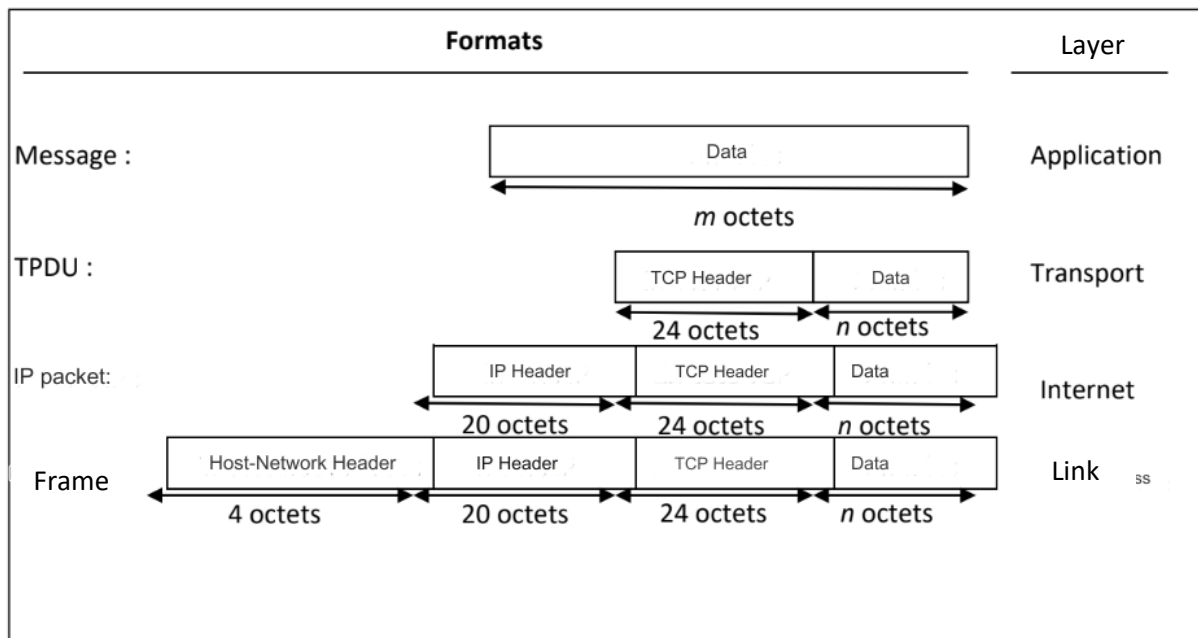
Review 2

Questions: Answer true or false, justifying your answer (in both cases).

1. In a bus topology, if one station fails, there will be an impact on the entire network.
2. The logical topology of a network depends on the physical topology.
3. According to the OSI model, a segment encapsulates a packet.
4. A **30-byte message** is transmitted from the application layer of system A to the application layer of system B. Each layer of the **OSI hierarchy** adds **5 bytes** of control information. The size of the message received by layer 1 of system B is **65 bytes**.
5. The control information placed before the data at the time of its encapsulation is called a capsule.
6. According to the OSI model, a frame is generated by the transport layer.

Exercise 1:

The TCP/IP layered structuring model is designed in four layers.



- 1) The Transport layer breaks the message into segments of maximum size 65536 bytes including a 24-byte TCP header. If the message (application level) is composed of 1024000 bytes:
 - a. What is the number of segments?
 - b. What is the size in bytes of each segment? Specify the size of the last segment.
2. At the Internet level a 20-byte header is added and at the Network Access level a 4-byte header is added.
 - a. How many frames are sent on the network?
 - b. What is the size in bytes of each frame? Specify the size of the last frame.
3. What is the total size in bytes of the data actually transmitted over the network?